

MEDIA & LINKS



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Using Images in Web Design

- **JPEG:** Best for photos, lossy compression
- **PNG:** Supports transparency
- **GIF:** Animation support
- **SVG:** Scalable vector graphics

Example:

```

```

```

```

- The **src** attribute shows the location or file name of the image. Here, the browser tries to display an image file named campus.jpg.
- The **alt** attribute provides an alternative text description of the image. This text is useful if the image cannot be loaded, and it is also important for accessibility because screen readers read this text aloud.
- The **width** attribute sets the width of the image to 400 pixels.
- The **height** attribute sets the height of the image to 250 pixels.
- The **title** attribute adds extra information about the image. In many browsers, this text appears when the user moves the mouse over the image.

In-Class Exercise - 1

1. Create a basic HTML document structure.
2. Add a heading above the image.
3. Insert an image using the **** tag.
4. Use and test the following **** attributes:
 - src
 - alt
 - width
 - height
 - title
5. Write meaningful values for each attribute.

Adding Videos to Webpages

- Formats: MP4, WebM, Ogg

Example:

```
<video width='600' controls>
```

```
  <source src='video.mp4' type='video/mp4'>
```

```
</video>
```

Alternative: Embed YouTube/Vimeo:

```
<iframe src='https://www.youtube.com/embed/xyz123'></iframe>
```

```
<video width="600" height="340" controls autoplay  
muted loop poster="cover.jpg">  
  <source src="sample-5s.mp4" type="video/mp4">  
  Your browser does not support the video tag.  
</video>
```

- **width:** Sets the video width.
- **height:** Sets the video height.
- **controls:** Shows the default video controls.
- **autoplay:** Starts the video automatically when the page loads. In many browsers, autoplay usually works only when the video is muted.
- **muted:** Starts the video without sound.
- **loop:** Repeats the video automatically after it ends.
- **poster:** Displays an image before the video starts playing.
- **src:** Defines the location of the video file.
- **type:** Defines the MIME type of the video file, such as video/mp4.

In-Class Exercise - 2

1. **Using your HTML document:**
2. Add a heading above the video.
3. Insert the video code into the `<body>`.
 - <https://samplelib.com/sample-mp4.html>
4. Add and test the following attributes:
 - width
 - height
 - controls
 - muted
 - loop
 - poster
5. Add a fallback message inside the `<video>` tag for unsupported browsers.

Using Audio in Web Design

- Formats: MP3, WAV, Ogg

Example:

<audio controls>

<source src='audio.mp3' type='audio/mpeg'>

</audio>

Best Practice: Avoid auto-playing sounds.

```
<audio controls autoplay muted loop>  
  <source src="sample-3s.mp3" type="audio/mpeg">  
  Your browser does not support the audio tag.  
</audio>
```

- **controls:** Shows the default audio controls.
- **autoplay:** Starts the audio automatically when the page loads. In many browsers, autoplay usually works only when audio is muted.
- **muted:** Starts the audio without sound.
- **loop:** Repeats the audio automatically after it ends.
- **src:** Defines the location of the audio file.
- **type:** Defines the MIME type of the audio file, such as audio/mpeg.

In-Class Exercise - 3

1. **Using your HTML document:**
2. Add a heading above the audio player.
3. Insert the audio code into the `<body>`.
4. Add and test the following attributes:
 - `controls`
 - `muted`
 - `loop`
 - `autoplay`

Favicon – Website Icon

- Small icon in the browser tab
- Enhances brand identity

Example:

```
<link rel='icon' type='image/png' href='favicon.png'>
```

Tools: Favicon.io, RealFaviconGenerator

```
<link rel="icon" type="image/png" sizes="16x16" href="
favicon-16x16.png">
```

-
- The **<link>** tag is used to connect the HTML document to an external resource inside the **<head>** section.
 - The **rel** attribute means relationship. Here, it tells the browser that the linked file is the icon of the webpage.
 - The **type** attribute specifies the file type or MIME type. In this example, the favicon is a PNG image.
 - The **href** attribute gives the path or file name of the favicon file. Here, the browser looks for a file named favicon.png.

In-Class Exercise - 4

1. Using your HTML document:
2. Add a page title inside the <title> tag.
3. Insert a favicon for your web page (inside the <head> section): <https://favicon.io/>
4. Save an image file named **favicon.png** in the same folder as your HTML file.
5. Open the page in the browser and check whether the favicon appears in the browser tab.

Hyperlinks & Navigation

- Internal
- External
- Email
- Phone Links

Examples:

`<a href='https://example.com' target='_blank'>Visit Website</a>`

`<a href="mailto:info@example.com">Send an Email</a>`

`<a href="tel:+1234567890">Call Us</a>`

Best Practice: Use descriptive link text.

```
<a href="https://www.istanbul.edu.tr" target="_blank"
title="Visit Istanbul University">Istanbul University
</a>
```

- The **<a>** tag is used to create a hyperlink in HTML.
- The **href** attribute stands for Hypertext Reference. It specifies the destination URL of the link.
- The **target** attribute defines where the link will open. **_blank** means the link opens in a new tab or window.
- The **title** attribute provides extra information about the link. In many browsers, it appears as a tooltip when the mouse moves over the link.
- The **<a>** tag has a clickable text «**Istanbul University**» shown to the user.

In-Class Exercise - 5

1. **Using your HTML document:**
2. Add a heading such as “Hyperlink Practice”.
3. Add an external link to a university or department website.
4. Add a link that opens in a new tab.
5. Add a link with a title attribute.
6. Add an email link using mailto.

 &

This code creates an ordered list.

Each tag defines one **list item**.

The items are shown with numbers.

```
<ol>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ol>
```

This code creates an unordered list.

Each tag defines one **list item**.

The items are shown with bullet points.

```
<ul>
  <li>Open the browser</li>
  <li>Write the website address</li>
  <li>Press Enter</li>
</ul>
```

In-Class Exercise - 6

1. **Using your HTML document:**
2. Add a heading such as «My Favorite Technologies».
3. Create an unordered list with at least five items.
4. Use `<li>` for each list item.
5. Add a short paragraph above the list explaining what the list shows.
6. Add another unordered list about a different topic, such as: favorite websites, courses this semester, programming languages, daily digital tools etc.

Challenge Task:

- Create a nested unordered list.
- Add a hyperlink inside one list item.
- Add an image inside one list item.
- Create one `<ul>` and one `<ol>` on the same page and compare them.

Navigation with <nav> Element

The **<nav>** tag is used to create navigation menus (main menu, side menu, footer menu, etc.) on a website. It is mainly used for linking to other pages, sections, or external links for easy navigation.

The <nav> tag is a **semantic HTML element**. This means it gives meaning to the content. It tells the browser and developers that this part of the page is used for navigation.

It is commonly placed at the top of a webpage in a sidebar or sometimes in the footer.

<header> tag creates a **header section** for a webpage. A header usually contains the **main title**, logo, introduction, or navigation elements.

```
<header>
  <h1>My Website</h1>
  <p>Welcome to our homepage</p>
</header>
```

In-Class Exercise - 7

1. **Using your HTML document:**
2. Add a heading such as «Website Navigation Example».
3. Add a `<nav>` section below the heading.
4. Inside `<nav>`, create at least four links using `<a>`.
5. Use example page names such as:
 - `index.html`
 - `about.html`
 - `services.html`
 - `contact.html`

Challenge Task:

- Add separators such as `|` between the links.
- Put the navigation menu inside a `<header>` section.
- Create a second `<nav>` for footer links.

Best Practices for Using Media & Links

- ✓ Optimize media for faster loading
- ✓ Use alt text for images
- ✓ Lazy loading improves performance
 - Lazy loading means waiting to render web content until the user or the browser needs it. It can help speed up webpage load times.
- ✓ Host large files externally
- ✓ Ensure proper navigation with structured links

References

Felke-Morris, T. (2017). *Web development and design foundations with HTML5* (8th Edition). Pearson, USA. ISBN 978-0-13-432275-9.

Robbins, J. N. (2012). *Learning web design: A beginner's guide to HTML, CSS, JavaScript, and web graphics*. O'Reilly Media, Inc. ISBN: 978-1-449-31927-4.

W3Schools HTML Tables: https://www.w3schools.com/html/html_tables.asp



Thanks!

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